

Biology for engineers

2 marks questions

- 1: What are biomolecules
- 2: Define monomers and polymers
- 3: What are amino acids
- 4: What are nucleotides
- 5: What is secondary structure of protein
- 6: Define enzymes
- 7: What is a nucleotide
- 8: Define gene
- 9: What is an allele
- 10: Define dominance
- 11: What is a single-gene disorder
- 12: Give one example of an autosomal recessive disorder
- 13: What is the role of meiosis in inheritance
- 14: Name any one stop codon
- 15: What is a nucleotide
- 16: Define Brownian motion
- 17: Who was Robert Brown
- 18: What is meant by science and engineering
- 19: Define unicellular organism
- 20: What are prokaryotes
- 21: Define autotrophs
- 22: What is ammonotelic excretion
- 23: What is molecular taxonomy
- 24: Name the three domains of life

5 marks question

1. Describe modes of nutrition: autotrophic, heterotrophic, and lithotrophic
2. Explain classification based on cellular organization and habitat
3. Explain levels of protein structure: primary, secondary, tertiary, and quaternary
4. Explain the classification of enzyme
5. Describe single-gene disorders in humans with examples
6. Explain the concept of complementation in human genetics with suitable examples
7. Write a short note on autosomal dominant and recessive disorders
8. Explain the double helix model of DNA
9. Explain the interrelationship of biology with physics, chemistry, and mathematics
10. Differentiate between science and engineering with examples

10 MARKS QUESTION

1. Explain historical biological observations with reference to Brownian motion
And thermodynamics
2. Describe classification in biology: cellularity, ultrastructure, and nitrogen
Excretion
3. Give a detailed account of biomolecules as the chemical basis of life, including
carbohydrates, proteins, and nucleic acids
4. Write a brief note on glycolysis
5. Explain Mendel's laws of inheritance with suitable examples
6. Explain DNA as the genetic material with experimental evidence

